CS3230 News (Spring 2015)

NUS National University of Singapore
School of Computing

CS3230 News:
Week 3, 28-Jan-2015

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Lecture on Week 6 (CNY week)

Proposal 0:
Keep Lecture 6 as scheduled in Week 6 on 18-Feb (eve of CNY)

Proposal 2:
Make-Up Lecture in Week 5 Fri, 13-Feb (6:30—8:30pm), LT1?

Prop 2: __28_____ **Prop 0: __>30_____**

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Help Session

Help Session for CS3230
On Math for Analysis of Algorithms

Time: Sat, 31-Jan-2015, 2 – 4pm.
Venue: SR3, COM1-02-12

Attendance is Optional

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Tentative stuffs to be covered…

Running Time of an Algorithm $T(n)$:
Meaning, How to analyze, How to use AA Patterns?
How to bound running times (upper bound, lower bound)?
How come we are obsessed with n, n^2, n^3, etc? 2^n, lg n (base 2)?

S1. Modified from Problem 3-3, pp 61-62 of [CLRS] [Sorting out order of growth rates]
Rank the following functions in *increasing order of growth*; that is, if function $f(n)$ is *immediately* before function $g(n)$ in your list, then it should be the case that $f(n)$ is $O(g(n))$.

$g_1(n) = n!$ $g_2(n) = 5n^2(\lg n)+14$ $g_3(n) = n^{\lg n}$ $g_4(n) = 4n^{0.5}$
$g_5(n) = 2^{2(\lg n)}$ $g_6(n) = 3(\lg n)^2$ $g_7(n) = 2^n$ $g_8(n) = n^2(\lg \lg n)$

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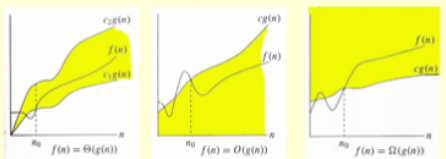
Tentative stuffs to be covered…

The O-notation, $T(n) = O(f(n))$: (also $\Omega, \Theta, o, \omega,$)

Why use O-notation for Running Time,
Why ignore constant factors? Why $n \geq n_0$?
Why funny definition? Why c, and n_0? How to find them?
Other (easy) methods to prove O, Ω, Θ
Why so many theorem?

Relations Between Θ, O, Ω



Theorem are Magic Potions.

Sum, Product Rule (Thm)

Polynomial Theorem

Limit Theorem

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Tentative stuffs to be covered…

How do we do wierd summation, estimation?

summing simple power-series,
estimation of sums using integrals

$$T(n) = n^2\left(1 + \tfrac{5}{16} + \left(\tfrac{5}{16}\right)^2 + \left(\tfrac{5}{16}\right)^3 + \cdots\right)$$

$$= n^2\left(\frac{1}{(1-\frac{5}{16})}\right) = \tfrac{16}{11}n^2$$

$x = \frac{5}{16}$

$$\ln(n+1) \leq \sum_{k=1}^{n} \frac{1}{k} \leq (\ln n) + 1$$

$$\sum_{k=1}^{n} \frac{1}{k} \geq \int_1^{n+1} \frac{1}{x}\,dx = [\ln x]_1^{n+1} = \ln(n+1)$$

$$\sum_{k=2}^{n} \frac{1}{k} \leq \int_1^{n} \frac{1}{x}\,dx = [\ln x]_1^{n} = (\ln n)$$

$$\Rightarrow \sum_{k=1}^{n} \frac{1}{k} \leq (\ln n) + 1$$

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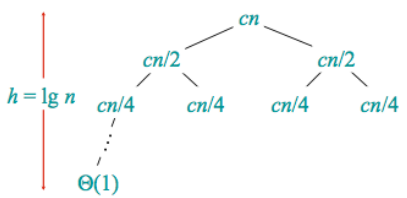
How did the (lg n) sneak into A.A.

MERGE-SORT $A[1 \,..\, n]$
1. If $n = 1$, done.
2. Recursively sort $A[\,1 \,..\, \lceil n/2 \rceil\,]$ and $A[\,\lceil n/2 \rceil + 1 \,..\, n\,]$.
3. "***Merge***" the 2 sorted lists.

Solve $T(n) = 2T(n/2) + cn$, where $c > 0$ is constant.



Find an element in a sorted array:

1.Divide: Check middle element.

2.Conquer: Recursively search 1 subarray.

3.Combine: Trivial.

Example: Find 9

3 5 7 8 9 12 15

Wouldn't it be easier if they write

for $k = 1$ to (lg n) do
do something

But NO.

Need to "discover" the hidden (lg n)

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